

Problem

A balanced delta-connected source has phase voltage

$$\mathbf{V}_{ab} = 880 \angle 30^\circ \text{ V}$$

and a positive phase sequence. If this is connected to a balanced delta-connected load, find the line and phase currents. Take the load impedance per phase as

$$60 \angle 30^\circ \Omega \text{ and line impedance per phase as } 1 + j1 \Omega.$$

Step-by-step solution

Step 1 of 7

Consider the circuit diagram shown in Figure 1.

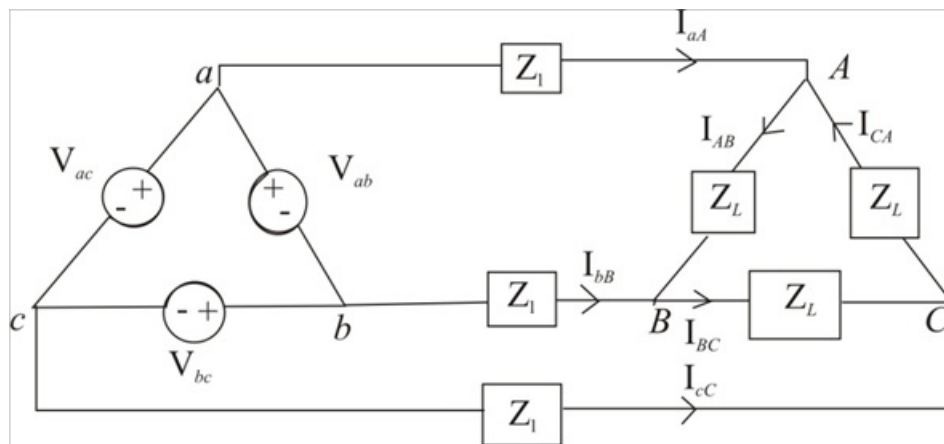


Figure 1

Step 2 of 7

Apply Kirchhoff's voltage law in Loop-1.

$$\begin{aligned} V_{ab} &= I_a Z_l - I_b Z_l + I_{AB} Z_L \\ &= (I_a - I_b) Z_l + I_{AB} Z_L \end{aligned}$$

In the positive sequence, the current I_b will be $I_a \angle -120^\circ$. The value of the current I_{AB} will be $\frac{I_a \angle 30^\circ}{\sqrt{3}}$ since, I_{AB} is a phase current.

$$V_{ab} = (I_a - I_a \angle -120^\circ) Z_l + \left(\frac{I_a}{\sqrt{3}} \angle 30^\circ \right) Z_L$$

Substitute $880 \angle 30^\circ$ V for V_{ab} , $1 + j1 \Omega$ for Z_l and $60 \angle 30^\circ \Omega$ for Z_L .

$$\begin{aligned} 880 \angle 30^\circ &= (I_a - I_a \angle -120^\circ)(1 + j1) + \left(\frac{I_a}{\sqrt{3}} \angle 30^\circ \right) (60 \angle 30^\circ) \\ &= (1.732 I_a \angle 30^\circ)(1.414 \angle 45^\circ) + (34.64 \angle 60^\circ) I_a \\ &= (1.732 \angle 30^\circ)(1.414 \angle 45^\circ) I_a + (34.64 \angle 60^\circ) I_a \\ &= (2.449 \angle 75^\circ + 34.64 \angle 60^\circ) I_a \end{aligned}$$

$$880 \angle 30^\circ = (37.01 \angle 60.98^\circ) I_a$$

$$I_a = \frac{880 \angle 30^\circ}{37.01 \angle 60.98^\circ}$$

$$I_a = 23.78 \angle -30.98^\circ \text{ A}$$

Therefore, the value of the line current I_a is $23.78 \angle -30.98^\circ \text{ A}$.

Step 3 of 7

Calculate the value of the current I_b .

$$I_b = I_a \angle -120^\circ$$

Substitute $23.78 \angle -30.98^\circ \text{ A}$ for I_a .

$$\begin{aligned} I_b &= (23.78 \angle -30.98^\circ)(1 \angle -120^\circ) \\ &= 23.78 \angle -150.98^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_b is $23.78 \angle -150.98^\circ \text{ A}$.

Step 4 of 7

Calculate the value of the current I_c .

$$I_c = I_a \angle 120^\circ$$

Substitute $23.78 \angle -30.98^\circ \text{ A}$ for I_a .

$$\begin{aligned} I_c &= (23.78 \angle -30.98^\circ)(1 \angle 120^\circ) \\ &= 23.78 \angle 89.02^\circ \text{ A} \end{aligned}$$

Therefore, the value of the line current I_c is $23.78 \angle 89.02^\circ \text{ A}$.

Step 5 of 7

Calculate the value of the phase current I_{AB} .

$$I_{AB} = \frac{I_a \angle 30^\circ}{\sqrt{3}}$$

Substitute $23.78 \angle -30.98^\circ$ A for I_a .

$$\begin{aligned} I_{AB} &= \frac{(23.78 \angle -30.98^\circ)(1 \angle 30^\circ)}{\sqrt{3}} \\ &= \frac{(23.78 \angle -0.98^\circ)}{\sqrt{3}} \\ &= 13.73 \angle -0.98^\circ \text{ A} \end{aligned}$$

Therefore, the value of the phase current I_{AB} is $13.73 \angle -0.98^\circ$ A.

Step 6 of 7

Calculate the value of the current I_{BC} .

$$I_{BC} = I_{AB} \angle -120^\circ$$

Substitute $13.73 \angle -0.98^\circ$ A for I_{AB} .

$$\begin{aligned} I_{BC} &= (13.73 \angle -0.98^\circ)(1 \angle -120^\circ) \\ &= 13.73 \angle -120.98^\circ \text{ A} \end{aligned}$$

Therefore, the value of the phase current I_{BC} is $13.73 \angle -120.98^\circ$ A.

Step 7 of 7

Calculate the value of the current I_{CA} .

$$I_{CA} = I_{AB} \angle 120^\circ$$

Substitute $13.73 \angle -0.98^\circ$ A for I_{AB} .

$$\begin{aligned} I_{CA} &= (13.73 \angle -0.98^\circ)(1 \angle 120^\circ) \\ &= 13.73 \angle 119.02^\circ \text{ A} \end{aligned}$$

Therefore, the value of the phase current I_{CA} is $13.73 \angle 119.02^\circ$ A.